Cases in Biochemistry and Genetics

Note: This reference document contains the content of the Canvas page for "Cases in Biochemistry and Genetics" with minor modifications for presentation in a document format. Refer to this session's Canvas page for the full interactive experience or if any hyperlinks do not function properly.

Overview

In the materials on this page, you will find some things that you have never seen before and some things that are revisiting topics you have seen before. The first section is about the inheritance of mitochondrial disease, with definitions of heteroplasmy and homoplasmy, which is then used to explain why mitochondrial disease is variable and not always as easy to recognize in a pedigree as the textbook version we talked about in semester 1. The second section revisits porphyrias, diseases that sometimes get overlooked clinically. The third section will revisit the coagulation cascade and hemophilias, with the intent of also revisiting X-linked recessive inheritance.

Learning Objectives

By the end of this session, you will be able to meet the following learning objectives:

1. Contrast mitochondrial and nuclear DNA structure, replication, and inheritance; include in the description heteroplasmy and the model for mild vs severe disease.
2. Describe the genetic etiology, pathophysiology, clinical presentation, and disease progression of mitochondrial encephalopathy with lactic acidosis and stroke-like episodes (MELAS).
3. Describe the pathway of heme synthesis including regulatory mechanisms and the difference between enzyme isoforms in precursors for red blood cells versus other cells.
4. Compare and contrast acute intermittent porphyria, porphyria cutanea tarda, and erythropoietic protoporphyria (with X-linked protoporphyria).
5. Describe the coagulation cascade including extrinsic, intrinsic, and common pathways.
6. Describe the common starting point of clot formation and how the signal becomes amplified to produce sufficient fibrin for clot formation.
7. Describe negative regulators of the coagulation cascade.
8. Describe the cause of hemophilia A, B, and C.

Preparation Content

For this session, content has been sectioned into separate sections. Visit these sections below to review content for each topic.

Cases in Biochemistry and Genetics: Part 1 of 3 – Heteroplamy

Overview

You should be fully acquainted with mitochondria from previous modules. As a reminder, this is where the TCA cycle and electron transport chain produce energy, fatty acids are broken down in beta-oxidation, and other roles include for example being intimately involved in apoptosis.

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Learning Objectives

This section covers the following learning objectives:

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2. Describe the genetic etiology, pathophysiology, clinical presentation, and disease progression of mitochondrial encephalopathy with lactic acidosis and stroke-like episodes (MELAS).

Mitochondrial DNA

One reason researchers think mitochondria derive from bacteria is their DNA structure and replication pattern. Recall that the human genome contains 22 pairs of autosomes and one pair of sex chromosomes with one maternal and one paternal homologue. The genome codes for approximately 25,000 genes, the chromosomes are linear, and they only replicate shortly before the cell undergoes mitosis.

In contrast, mitochondria contain one circular piece of DNA—mtDNA—stored in the mitochondrial matrix. It is polyploid, with up to 10 copies of mtDNA per mitochondrion. This DNA can replicate independently of cell or mitochondrial division. Mitochondria are maternally inherited, so disorders caused by mutations in mtDNA are only passed down by the mother.

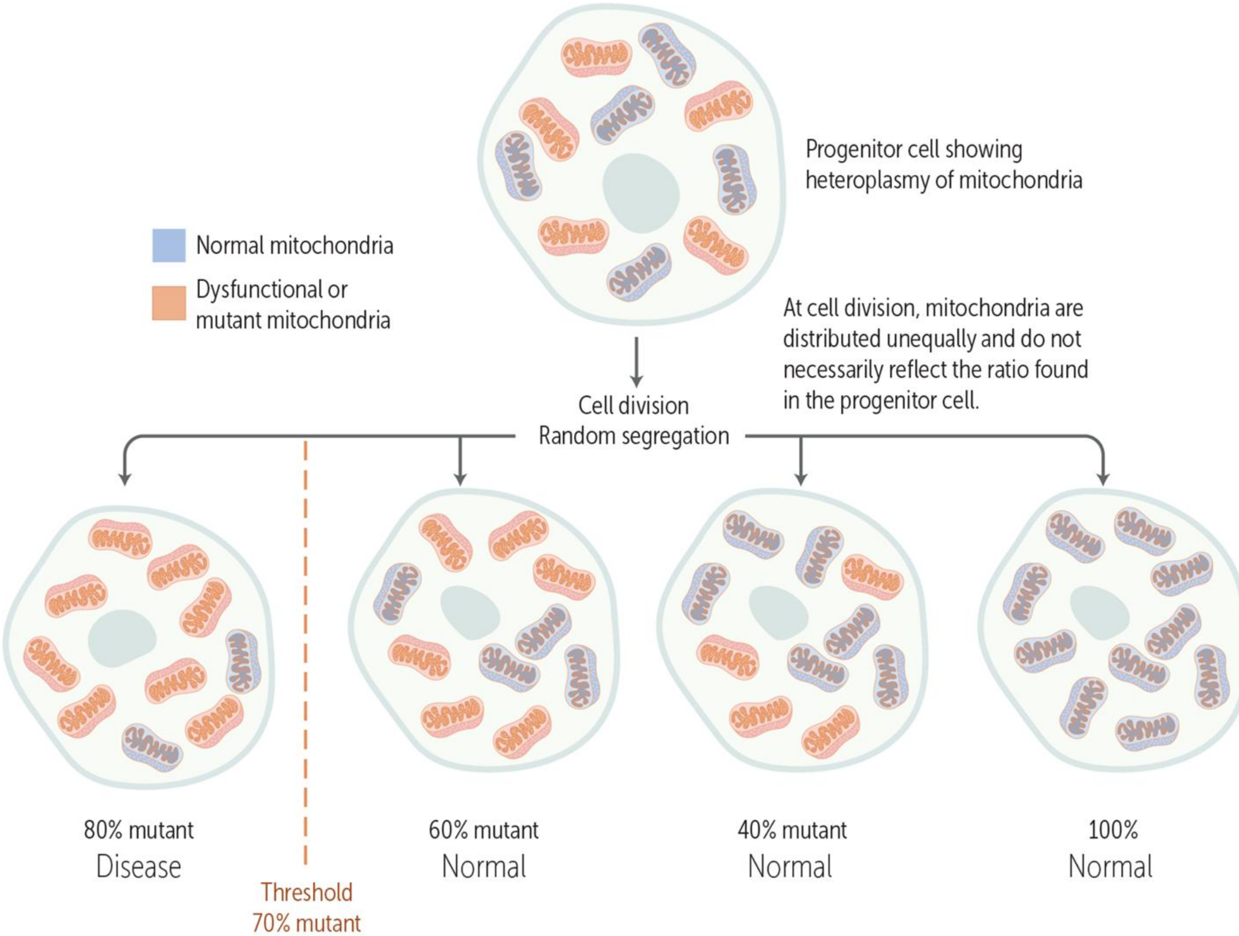
Mitochondria contain approximately 1700 proteins, only 13 of which are encoded by mtDNA. All 13 are part of the oxidative phosphorylation pathway. MtDNA also codes for 22 transfer RNAs (tRNAs) and two ribosomal RNAs, which are used for translation in the cytosol of the mitochondria, for a total of 37 genes. This is pretty tiny compared with the human genome! One drawback to this minuscule genome is the lack of DNA repair genes. The DNA repair proteins coded by nuclear DNA primarily return to the nucleus, limiting mutation surveillance and DNA repair in the mitochondria. Combined with the large amount of ROS generated from oxidative phosphorylation, mtDNA acquires mutations at about 10 times the rate of nuclear DNA. Mitochondrial changes due to the high mutation rate may be implicated in ageing.

Replicative Segregation and Threshold of Disease

Because fission and mtDNA replication are uncoupled from each other and from the cell cycle, mitochondria undergo **replicative segregation**. Essentially, mtDNA randomly segregates between mitochondria during fission, and mitochondria randomly segregate to the two daughter cells after mitosis. This has implications for disease manifestation.

Remember how nuclear DNA has a wild-type and variant designation for mutations? Something similar occurs with mtDNA, but we use different terms because mtDNA can replicate at any time, multiple copies of mtDNA can be present in one mitochondrion, and a cell may contain up to 1000 mitochondria: the traditional terms just don't cover all the possibilities. When all of the mtDNA material has the same sequence (all normal or all mutant), it's called homoplasmy (homo = same, plasmy = plasmid). When there is variation in the mtDNA sequence (some normal and some variant), it's called heteroplasmy (hetero = different). Heteroplasmy can manifest within one mitochondrion, a tissue, or an individual. We generally only consider heteroplasmy at the tissue or individual level as potentially clinically significant.

Recall that for genetic diseases arising from mutations in nuclear DNA, we classify them as recessive or dominant depending on the inheritance pattern, and some disorders can manifest with only one dysfunctional or missing protein, whereas other disorders manifest only when no functional proteins are present. Now let's expand this concept to mtDNA and mitochondrial genetic disorders. With up to 10 mtDNA copies per mitochondrion, and up to 1000 mitochondria per cell in different tissues, the amount of mtDNA genetic variation in a person can be quite large. Just like some genetic diseases need one or both copies of a gene lost to show signs and symptoms, mitochondrial disorders only manifest if enough copies of mtDNA containing a mutated gene are within a tissue. This is called the **threshold model of disease**.



Threshold model of disease. Image credit: ScholarRx, used with permission.

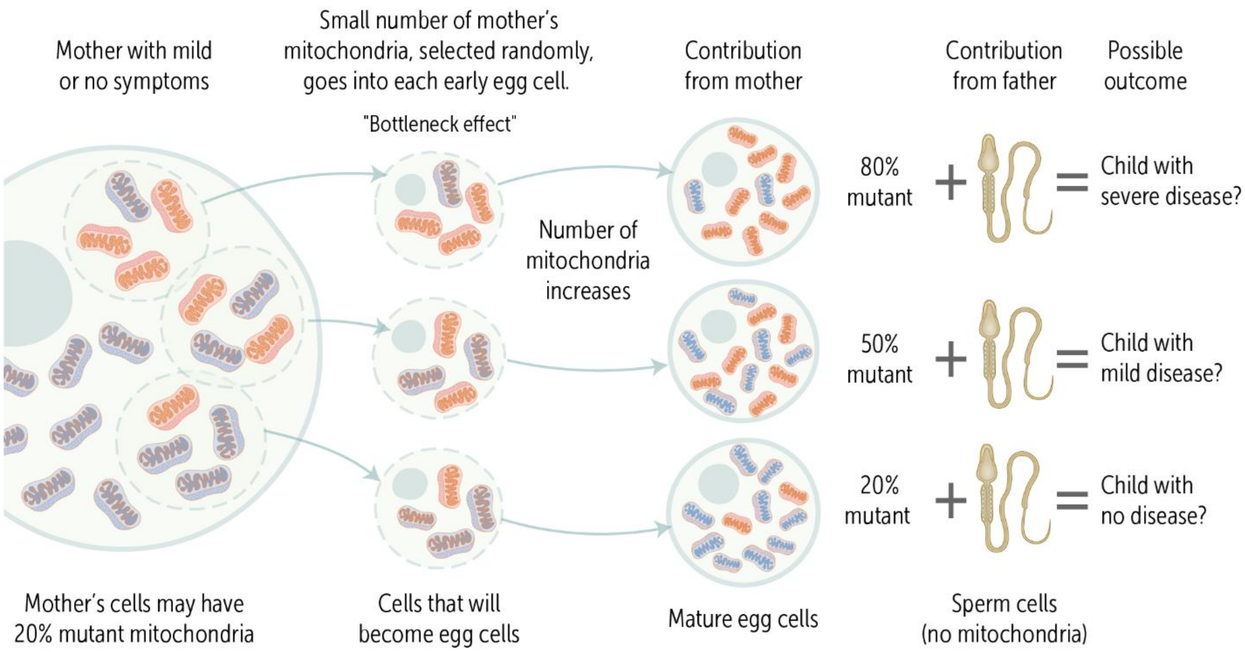
Replicative segregation means that each daughter cell gets a random number of healthy and mutated mtDNA distributed among a random set of mitochondria. Only when a sufficient number of mitochondria contain enough dysfunctional genetic material to affect cellular functions do we see manifestations of mitochondrial disorders. This principle acts just the same whether the cell undergoes mitosis or meiosis, so this also applies to gametes. As you have seen, the more important gamete regarding mitochondria is the oocyte.

We are here examining maternally inherited mitochondrial disorders. All children of a mother with a high enough mutation burden, or a mother who manifests a mitochondrial disorder, are at risk of inheriting these disorders. It's extremely difficult to calculate the probability of a child inheriting a mitochondrial disorder:

- Replicative segregation makes it hard to predict the mutation burden of offspring.
- Calculating the percentage of heteroplasmy in an individual is challenging.
- Mutation burden varies between tissues, so the percentage of heteroplasmy in muscle, for example, may not reflect the percentage of heteroplasmy in the oocytes.

- Diagnosing some mitochondrial disorders is challenging because of vague symptoms, and women may not be diagnosed until they have a child who manifests more severe symptoms than the mother does.

The probability that a woman will have a child with a mitochondrial disorder is dependent on the percentage of variant mtDNA, the threshold of disease, and the bottleneck effect, in which a random portion of the mother's mitochondria contributes to the mature oocyte. Thus, whether a child manifests a mitochondrial disorder is dependent on the percentage of variant mtDNA inherited from the mother. For example, a healthy mother with a 20% mutation burden may or may not have a child with a mitochondrial disorder (see image below). Notice that even though the sperm contains a lot of mitochondria, these are not contributed to the zygote, only the nucleus from the sperm enters.



A healthy mother with a 20% mutation burden may or may not have a child with a mitochondrial disorder.
Image credit: ScholarRx, used with permission.

Clinical Correlation

Reproductive options for women with mitochondrial disorders include using donor eggs, adoption, and potentially three-parent in vitro fertilization, using the nucleus from the parent's gametes and mitochondria from an egg donor.

What Are Mitochondrial Encephalomyopathies?

Enzymes for oxidative phosphorylation and tRNAs are the two main groups of molecules mtDNA codes for. You can imagine that defective oxidative phosphorylation will most significantly affect tissues with a high metabolic demand, such as the brain and skeletal muscle. Inherited encephalomyopathies are a group of disorders that present with neurological dysfunction and muscle weakness, plus a variety of other signs and symptoms depending on the affected gene. There are more than 200 oxidative phosphorylation proteins, of which mtDNA codes for 13, so inherited encephalomyopathies can present with a maternal inheritance pattern if the affected gene is coded by mtDNA or an autosomal recessive or X-linked recessive pattern if the affected gene is coded by nuclear DNA. Diseases caused by abnormal mitochondrial energy generation are the most common group of inherited metabolic disorders. Here we will focus on mitochondrial encephalopathy with lactic acidosis and stroke-like episodes (MELAS). Description of a couple of others — Leigh syndrome, and myoclonic epilepsy with ragged red fibers (MERRF) is available in a Brick on ScholarRx.

Mitochondrial Encephalopathy with Lactic Acidosis and Stroke-Like Episodes

MELAS is a relapsing-remitting, progressive neurological disorder that is purely maternally inherited.

Genetics

Most cases are due to a single point mutation in the mitochondrial tRNALeu gene, which interferes with tRNA processing. The mature tRNA lacks precise and efficient codon recognition, resulting in premature protein generation, low rates of protein synthesis, and elongation termination.

At the biochemical level, this decreases ATP production (from both glucose and fat) due to oxidative phosphorylation complex deficiencies, primarily in complexes I and V (even though mitochondrial gene products are found in all five complexes). Low complex V activity decreases hydrogen ion transport so that resting membrane potential cannot be maintained and calcium sequestration cannot occur. In addition, abnormal ROS production can lead to further DNA damage or a decrease in ROS signaling molecules such as nitric oxide.

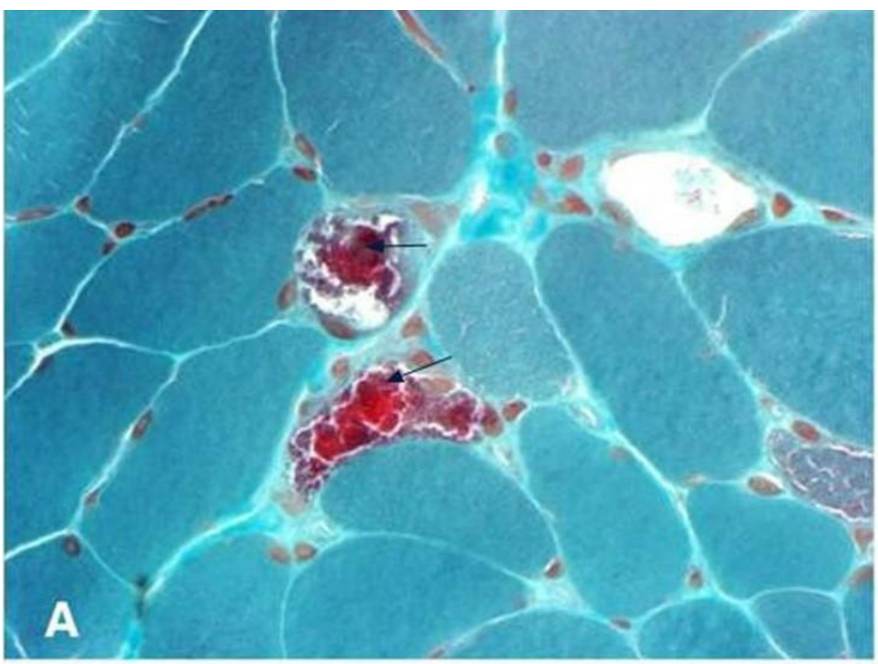
The mutation load significantly varies between tissues, with the highest burden in skeletal muscle and brain, and increases with age. Thus, heteroplasmy causes a significant variation in disease severity and presentation.

Presentation

MELAS presents with abrupt onset of stroke-like episodes, migraines, and seizures, usually in childhood or adolescence. Some affected individuals may have a history of muscle weakness. Stroke-like episodes often present with headache and vomiting, lasting from a few hours to a few days, and seizures. Transient hemiplegia or hemianopia may last for a few hours to several weeks after the episode. Abnormal metabolic activity results in increased lactate and pyruvate concentrations in blood and cerebrospinal fluid, especially after a stroke-like episode, but does not cause systemic metabolic acidosis. The stroke-like episodes lead to progressive neurological dysfunction, muscle weakness, aphasia, and dementia. Patients usually have a shortened lifespan, though this is dependent on disease severity.

Diagnosis

Imaging studies show diffuse cerebral or cerebellar atrophy, calcification lesions in the basal ganglia, or cortical and white matter lesions in the brainstem, basal ganglia, cerebellum, or temporo-parieto-occipital region. Episodes are classified as stroke-like because lesions in these areas do not include vascular territory and the lesion pattern changes faster than expected for ischemic stroke. Muscle biopsies commonly show abnormal size variation and activity of type 1 and type 2 fibers, ragged red fibers, and abnormal mitochondrial morphology in small blood vessels. Ragged red fibers are due to mitochondrial accumulation in the subsarcolemma of the muscle fiber.



Modified Gömöri trichrome stain. showing several ragged red fibers (arrowhead). Abu-Amero et al. Journal of Medical Case Reports 2009 3:77 doi:10.1186/1752-1947-3-77. Modifications made by CopperKettle. CC BY 2.0.

Summary

- Mitochondria are organelles responsible for many metabolic functions, regulation of apoptosis, calcium sequestration, and generation of secondary messengers.
- Mitochondria contain multiple copies of circular DNA (mtDNA), and each cell can contain up to 1000 mitochondria depending on energy needs.
- Mitochondria divide and mtDNA replicates independent of cellular mitosis.
- Replicative segregation means that mtDNA randomly segregates after mitochondrial division, and mitochondria randomly segregate in cellular mitosis or meiosis.
- A tissue or individual can be homoplasmic, with all mtDNA the same sequence, or heteroplasmic, with variation in mtDNA sequence.
- The threshold model of disease states that mitochondrial disorders arising from mutations in mtDNA only manifest if the percentage of variant mitochondria is high enough to affect function.
- Mitochondria are maternally inherited, so diseases that are due to mutations in mtDNA are passed from mother to children.
- Because metabolic enzymes in the mitochondria arise from both mtDNA and nuclear DNA, as a class, mitochondrial disorders can have autosomal recessive, X-linked recessive, or maternal inheritance.
- Mitochondrial encephalomyopathies are all characterized by abnormal energy generation, increased pyruvate and lactate, progressive central nervous system atrophy, and muscle weakness.
- MELAS is caused by a maternally inherited point mutation in tRNALeu, resulting in low expression and activity of oxidative phosphorylation complex I and V proteins.
- MELAS presents with stroke-like episodes starting in late childhood to adolescence and progresses to dementia, aphasia, and myopathy.

Cases in Biochemistry and Genetics: Part 2 of 3 - Porphyria

Overview

In this section we are revisiting some of the material you learned in the HOII module. As you should remember, porphyrias are diseases caused by accumulating precursors in the production of heme, and the cause of accumulation is deficiency of the enzyme that should use the precursor as substrate. The biochemical reactions involved in producing heme is summarized in the below figure. Please notice that the first and the last few steps of the synthesis takes place in mitochondria, while the middle steps happen in cytosol. Please notice that the information given here include the entire pathway and all the porphyrias; but notice that in the second part of this section, we will concentrate on some more common (less rare) porphyrias.

You have in the past learned a rule of thumb that enzyme deficiencies more frequently than not are inherited as autosomal recessive conditions and a couple meet that expectation and one is X-linked recessive. However, some porphyrias are exceptions to the rule, in that they show autosomal dominant inheritance but with reduced penetrance.

Please also recognize that one good source of patient information is the patient organization, the American Porphyria Foundation.

Some content and images for this lecture are adapted from ScholarRx and used with permission.

Learning Objectives

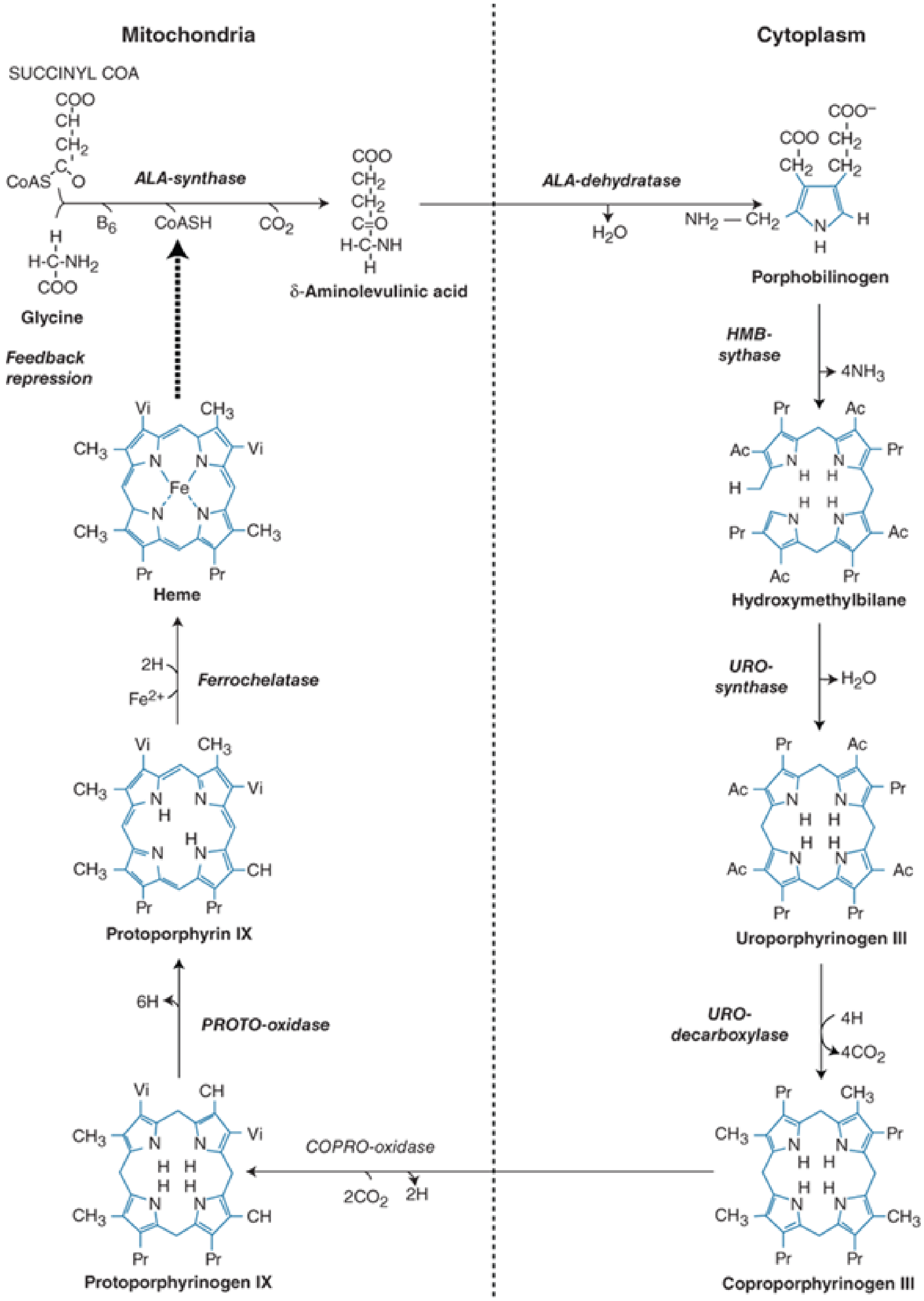
This section covers the following learning objectives:

3. Describe the pathway of heme synthesis including regulatory mechanisms and difference between enzyme isoforms in precursors for red blood cells versus other cells.
4. Compare and contrast acute intermittent porphyria, porphyria cutanea tarda, and erythropoietic protoporphyria (with X-linked protoporphyria).

Production of Heme and Intro to Porphyrias

In this figure, notice that deficiency in the last several enzymes, from URO synthase, will result in photosensitivity due to formation of complex rings. These rings will tend to distribute in the body including skin, and there, interaction with light will lead to formation of reactive oxygen radicals. The reason that deficiency of URO synthase also leads to photosensitivity is that the precursor hydroxymethylbilane will spontaneously become a uroporphyrin analogue in the absence of enzyme activity.

The next thing to notice in the figure is that the end product, heme, is an inhibitor of the very first reaction in the process, performed by ALA synthase.

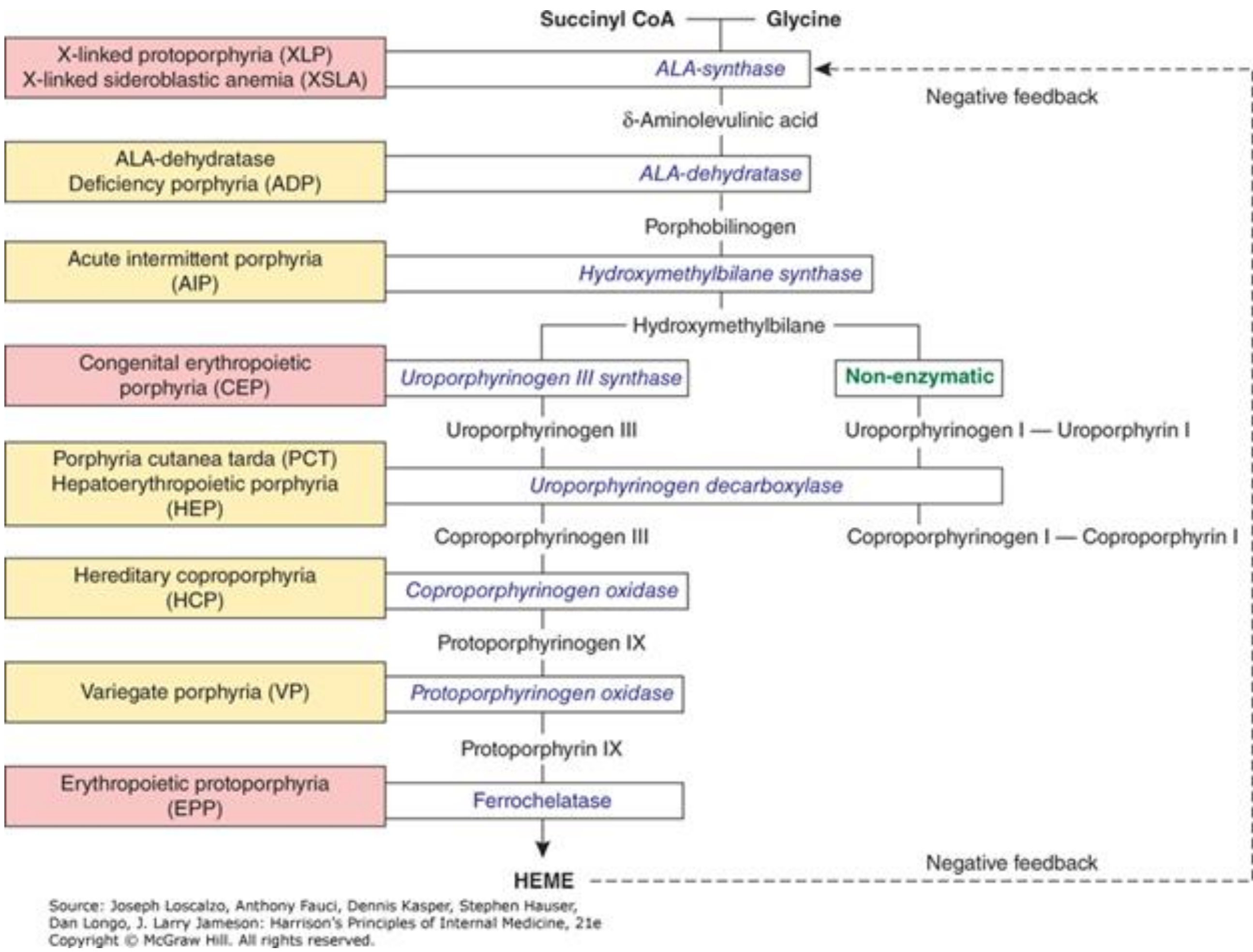


The heme biosynthetic pathway showing the eight enzymes and their substrates and products. Four of the enzymes are localized in the mitochondria and four in the cytosol. Image credit: McGraw Hill, used with permission.

The next thing to notice in the figure is that the end product, heme, is an inhibitor of the very first reaction in the process, performed by ALA synthase.

One thing you have heard before is that heme production is sensitive to lead poisoning. Lead inhibits the second step (ALA dehydratase) and the last step (ferrochelatase). Interestingly, in the last of these, a divalent ion is still found in the heme group, but it is zinc instead of iron when lead interacts with this system.

The next figure shows the enzymes involved in this pathway and the porphyrias associated with deficiency of them. As you can see, there is a classification of hepatic versus erythropoietic porphyrias. This is based on the origin of the intermediate that is causing symptoms, where it is produced.



The human heme biosynthetic pathway indicating in linked boxes the enzyme that, when deficient or overexpressed, causes the respective porphyria. Hepatic porphyrias are shown in yellow boxes and erythropoietic porphyrias in pink boxes. Image credit: McGraw Hill, used with permission.

Again, as a reminder, heme is produced in the red blood cell precursors for use in hemoglobin, but also in liver as part of producing for example cytochrome P450 enzymes and in most tissues for production of electron carriers in the electron transport chain.

The second frequently used classification of porphyrias is in acute vs chronic. This will be obvious when looking at the table below, where one is chronic and the other two are acute conditions.

The Three Most Common Porphyrias and Their Contrasting Presentations, Exacerbating Factors, and Approaches to Diagnosis and Treatment

Disease	Presenting Symptoms	Exacerbating Factors	Most Important Screening Tests	Treatment
Porphyria cutanea tarda (PCT)	Chronic, blistering skin lesions*	Iron; alcohol; smoking; estrogens; hepatitis C; HIV; halogenated hydrocarbons	Plasma or urine porphyrins¶	Phlebotomy; low-dose hydroxychloroquine or chloroquine
Acute intermittent porphyria (AIP)	NeurovisceralΔ	Drugs (mostly cytochrome P450-inducers); progesterone; dietary restrictionsΔ	Urinary porphobilinogen (PBG)Δ	Hemin; glucoseΔ
Erythropoietic protoporphyria (EPP)	Acute, nonblistering skin photosensitivity		Erythrocyte protoporphyrin (total and metal-free)	β-carotene; afamelanotide

Porphyria cutanea tarda (PCT) and acute intermittent porphyria (AIP) are distinct, but each shares features with some less common porphyrias (listed in the footnotes and discussed in detail in the overview of porphyrias topic in UpToDate).

β: beta.

* Also may occur in variegate porphyria (VP), less commonly in hereditary coproporphyria (HCP), and rarely in AIP with concomitant renal failure; much more severe in congenital erythropoietic protoporphyria (CEP) and hepatoerythropoietic porphyria (HEP).

¶ Also useful for diagnosis of variegate porphyria (VP), congenital erythropoietic porphyria (CEP), and hepatoerythropoietic porphyria (HEP).

Δ Features shared with hereditary coproporphyria (HCP), variegate porphyria (VP), and delta-aminolevulinic acid (ALA) dehydratase porphyria (ADP).

Further Presentation of Porphyrias

Acute Intermittent Porphyria

This is a condition that usually appears in heterozygotes for inactivating pathological variants of the gene for hydroxymethylbilane synthase. (A few homozygotes for pathological variants have been identified; they experience a more severe phenotype including onset usually during childhood). However, the number of people carrying a variant allele is much higher than the number of people showing the disease, indicating that one functional allele producing 50% protein activity is sufficient for normal phenotype most of the time. In conclusion, this is an autosomal dominant disease but with low penetrance.

The name of this condition includes the word "acute" to indicate that the condition is episodic in most patients, and this also explains why many carriers never progress to have a diagnosis. The factors that induce the attack include environmental, dietary, and steroid hormonal factors. This condition is more frequent in females, and contraceptives, other drugs, and low caloric intake can induce episodes. There is a large collection of drug information that should be considered when treating a person with acute porphyrias.

Why does low caloric intake influence this system? To understand this, we need to understand the difference between a steady state without symptoms and an acute episode. In steady state, the initiation of heme production is relatively low, and the amount of enzyme produced by one functional allele of hydroxymethylbilane synthase is enough to process the compounds made by previous steps in the pathway. An attack will happen when the initial step in the pathway has its activity upregulated beyond what the one allele of the enzyme can handle. During a strong limitation of caloric intake, such as during disease or attempts of losing weight, the person will begin relying on fat stores for caloric requirements. One hypothesis states that under normal conditions, fat acts as a buffer for fat soluble toxic compounds and that during caloric restriction some of these toxins will be released into the blood. Therefore, during caloric restriction, the liver will ready itself for needing to detoxify those compounds by producing extra cytochrome P450 enzymes, and therefore increase production of heme. The mechanism by which this regulation takes place seems to involve the peroxisome proliferator-activated receptor γ coactivator 1α (PGC-1α), a molecule that is increased during fasting and which seems to regulate the gene for ALA synthase.

Next, what about the effect of steroids? One finding is that the promoter for ALA synthase contains binding sites for the estrogen receptor and that in experimental systems, estrogen increases the production of this enzyme. Such an effect of estrogen could explain why attacks of AIP are rare in prepubertal children as well as why females are more likely to experience attacks. Another finding is that some metabolic products from steroids including e.g., from progesterone in contraceptives may have an even stronger effect on the levels of this enzyme.

At the clinical level, abdominal pain is the most common symptom. This and other symptoms are neurological and not inflammatory, indicating a direct effect of the porphyrin precursors on nerves. Nausea; vomiting; constipation; tachycardia; hypertension; central nervous system symptoms; pain in the limbs, head, neck, or chest; muscle weakness; sensory loss; dysuria; and urinary retention are characteristic but could also be due to other causes. In more severe attacks, there is peripheral neuropathy due to direct action on axons, not due to demyelination. Central nervous system symptoms can include depression, anxiety, sleep disturbances, and in severe cases seizures. Many drugs used for these types of symptoms in other people risk making the situation worse due to the drug interactions already mentioned. The severity of symptoms can vary significantly, and severely affected patients may have frequent attacks which together with the mental symptoms make the disease debilitating. Long term effects include kidney damage, hypertension, and a risk of liver carcinomas.

Once a case of porphyria is suspected, analysis of heme precursors in urine or serum is needed. In the case of AIP, the precursors would include both delta-aminolevulinic acid (delta-ALA) and porphobilinogen. Likely, a follow up test using DNA sequencing of the genes will follow. Treatments include pain relief and intravenous hemin (a heme analogue) and possibly glucose infusion in order to counter the effect of fasting. Other drug treatments were covered by pharmacology during the HOII module. In the worst cases, liver transplant would be recommended.

Knowledge Check Question

Question: Why would a heme analogue be given in acute intermittent porphyria?

Answer: Because heme is a feedback inhibitor of the first step in heme synthesis.

Porphyria Cutanea Tarda (PCT)

This condition appears when activity in the liver of the enzyme uroporphyrinogen decarboxylase (URO-D) falls below around 20% of normal activity. The cause of this can be difficult to discern. In 20% of cases, the patient has inherited one inactivated copy of the gene for the enzyme, but this in itself only leads to 50% reduction in activity and further reduction therefore is necessary before the disease appears. The remaining about 80% have no deficiency of the gene, but enzyme activity is still reduced. The explanation for the reduction seems to be appearance of an inhibitor in the hepatocytes, with **an oxidized derivative of uroporphyrinogen III or uroporphyrinogen I** being the most likely culprit. For completeness, the related condition of having inherited homozygosity for an inactivating variant is known as hepatoerythropoietic porphyria (HEP) and differs in having the low activity of the enzyme in all tissues, not only in liver.

Risk factors for development of the disease include above normal iron stores (for example due to hemochromatosis), elevated alcohol intake, estrogen supplementation, or infection with HIV or hepatitis C virus. Iron is a well known risk factor for oxidative stress, and most of the other risk factors seems to increase production of cytochrome p450 or in general challenge the integrity of the hepatocytes.

The disease is divided into types 1 and 2, where type 1 is also called sporadic and covers those patients without inherited URO-D variants, while type 2 is called familial because the person is heterozygote for a pathological variant. However, a person with type 2 does not have to have any family history of disease, because many heterozygotes do not develop disease.

Symptoms of PCT include a chronic, crusted blistering of sun-exposed areas as seen in the image. These will lead to scarring and hyperpigmentation, etc., which can be disfiguring. However, there are usually no neurological symptoms in this condition. Diagnosis will start with physical examination followed by analysis of **urine or plasma** for intermediates of this pathway, with **uroporphyrins the major** component detected.

Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e

Typical cutaneous lesions in a patient with porphyria cutanea tarda. Chronic, crusted lesions resulting from blistering due to photosensitivity on the dorsum of the hand of a patient with porphyria cutanea tarda. Image credit: McGraw Hill, used with permission.

A number of treatments can be attempted, of which phlebotomy is usually effective. If contraindicated, low levels of chloroquine or hydroxychloroquine can be used because they complex porphyrins and augment their excretion. If hepatitis C virus is involved, then antiviral drugs may be the best treatment.

Erythropoietic Protoporphyria (EPP) and X-linked Protoporphyria (XLP)

Two different underlying causes result in clinically very similar outcomes and are therefore treated together here. The condition results when the last precursor of heme (protoporphyrin IX) is elevated in the body of the patient. The cause of both conditions is genetic, one is autosomal recessive and the other X-linked.

As the name erythropoietic protoporphyria for the first of these indicates, this is a condition where the precursor is produced during erythropoiesis rather than in liver cells. The cause is reduced activity of ferrochelatase (FECH) which is caused by the patient being homozygote or compound heterozygote for variants in this gene – the inheritance pattern therefore is autosomal recessive. Most patients of this group have one inactivating mutation in one allele combined with a common allele of the gene which causes aberrant splicing and therefore reduced production of FECH (IVS3-48T/C) as the second allele. There have been reports that the low expression allele (48C) occurs with varying frequency in different populations, about 1% in people from western Africa, 10% in whites and 43% in Japanese.

The second condition, X-linked protoporphyria, is caused by an activating mutation in the gene encoding the erythroid-specific isoform of ALA synthase (ALAS2). Males with this condition are clinically very difficult to distinguish from patients with EPP, while female carriers have variable phenotype; manifestation of symptoms in carriers seems to be caused by skewed X-chromosome inactivation. The cause in this case for symptoms is still accumulation of protoporphyrin IX, seemingly because that is the step most easily overwhelmed when the initial step in the pathway is overly active. About 10% of the patients presenting with these symptoms have the X-linked form.

Symptoms of these conditions is centered on painful photosensitivity, usually without blistering. The onset is usually during childhood. When exposed to sun, within between <10 and 60 minutes there will be onset of sensations that can include tingling, burning, itching etc. If the person stays in the sun, there can be progression to a much more painful full-blown attack that lasts 2-5 days and which can be incapacitating. This attack is accompanied by redness and swelling in exposed areas and the pain is described as disproportionate to the physical changes. Over time, some relatively mild irreversible changes in the skin of exposed areas can happen. Up to ¼ of these patients have some changes in liver function, and in a few percent, this can develop to liver failure necessitating liver transplantation.

The protoporphyrin is cleared by the liver into bile, and as such there is a possibility for development of gallstones in these patients. This occurs in a minority, but if a child has gallstones, this should definitely be one of the causes investigated.

Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e

Erythema and edema of the hands due to acute photosensitivity in a 10-year-old boy with erythropoietic protoporphyria. Image credit: McGraw Hill, used with permission.

Confirmation of the diagnosis involves testing a blood sample for protoporphyrin IX (it will be carried by albumin); perhaps better to test erythrocytes as the molecules in plasma can also come from liver, and the hypothesis is a specific deficiency in the erythropoiesis. Because a few other conditions (including lead poisoning and iron deficiency) can also result in elevated protoporphyrin, it is important to test for whether the molecule contains a chelated ion. With the diseases described here, the photoporphyrin will be empty, while it will be complexed with zinc in the alternatives. A follow up DNA sequence of the involved genes is usually also done. Treatment will mainly rely on wearing protective clothing and recently, drugs that increase tanning without sun exposure are also being developed.

Cases in Biochemistry and Genetics: Part 3 of 3 - Clotting

Overview

The coagulation cascade is a complex series of enzymatic reactions essential for blood clotting, involving both intrinsic and extrinsic pathways that converge to form a stable fibrin clot. Negative regulators of this cascade, such as antithrombin, protein C, and protein S, help prevent excessive clot formation and ensure balance in the hemostatic process. Hemophilia, a genetic disorder typically caused by mutations in the genes encoding clotting factors VIII (Hemophilia A) or IX (Hemophilia B), results in impaired clot formation and prolonged bleeding.

Some content and images for this lecture are adapted from ScholarRx and used with permission.

Learning Objectives

This page covers the following learning objectives:

5. Describe the coagulation cascade including extrinsic, intrinsic, and common pathways.
6. Describe the common starting point of clot formation and how the signal becomes amplified to produce sufficient fibrin for clot formation.
7. Describe negative regulators of the coagulation cascade.
8. Describe the cause of hemophilia A, B, and C.

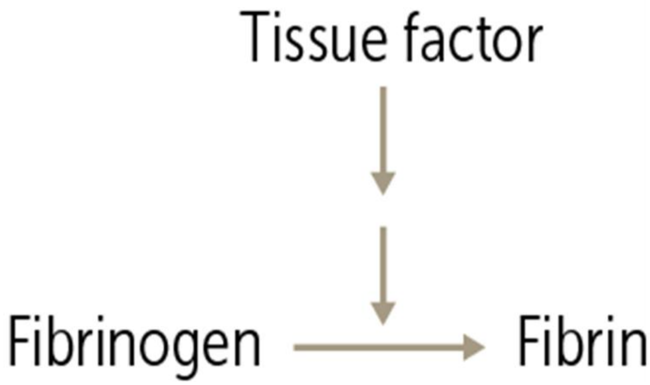
What Is the Coagulation Cascade?

As you hopefully are aware already, there are three steps in making a clot:

- Blood vessel constriction
- Platelet plug formation
- Coagulation

This text concentrates on the last step in clot formation, coagulation. This is the process that will ultimately split fibrinogen to fibrin, creating the "glue" that holds together the forming clot.

How is this critical fibrin "glue" formed? It comes from the clotting cascade, a biochemical process important enough to revisit. The components involved are proteins circulating in blood that activate each other in a cascade fashion (hence the name). The cascade starts when the membrane protein **tissue factor** (TF) is exposed to the blood, and the whole point of the resultant cascade is to **turn fibrinogen into fibrin**, to seal up the nice mushy platelet plug which was just made. If we wanted to really simplify it, we could draw the cascade like the figure below.



Fibrinogen to fibrin cascade. Image credit: ScholarRx

The steps filling out this basic cascade are the subject of this text. It's worth your time to really understand these processes because it will make learning about health, diseases, and drugs so much easier. Once you understand the cascade, you don't have to simply memorize how any number of processes may tip the balance in one direction or another; they will just make sense.

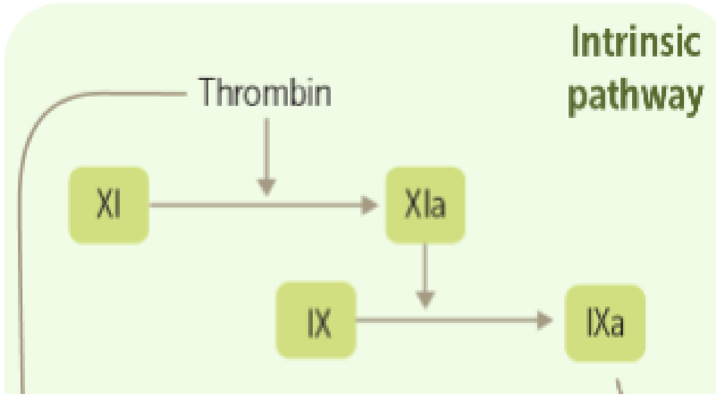
Knowledge Check Question

Question: What is the final purpose of the coagulation cascade?

Answer: The final purpose of the coagulation cascade is turning fibrinogen into fibrin, which forms a network that stabilizes the soft platelet plug.

What Are Coagulation Factors and Cofactors?

As you might have guessed, enzymes (or "factors") form the backbone of the coagulation cascade. Their main job is to activate other enzymes. Most of the coagulation factors are synthesized by the liver and circulate in the blood as **inactive enzyme precursors**. When the cascade is initiated, one activated factor cleaves and thereby activates the next factor, which again cleaves and activates the following factor. As each activated enzyme acts on multiple precursor molecules at the next level, this amplifies the signal. Each factor is named using a roman numeral, and the **active enzymes** are indicated by the lowercase "a" following the Roman numeral. The entire cascade is shown further down this page. The small figure here shows the first two steps of the intrinsic pathway of the coagulation cascade. From thrombin activating XI to XIa, and this in turn activating IX to IXa, one molecule of thrombin can in principle result in between 1,000 and 1,000,000 active copies of IXa.



Cascade Initiation. Image credit: ScholarRx, modified and used with permission.

Most of these enzymes are **serine proteases**, which means they hydrolyze peptide bonds using a serine residue at their target. The exception is factor XIII, a transglutaminase, which crosslinks and stabilizes fibrin polymers. In addition to the enzymes, there are a couple of cofactors which act by forming heterodimers with an enzyme to activate it.

Individual Factors

The coagulation factors with enzymatic activity include factors II, VII, IX, X, XI, XII, and XIII. When an "a" is placed after the factor name (eg, IIa), it means the factor has been cleaved into an **activated version** by another enzyme. For example, factor XIa cleaves factor IX into an activated factor IXa.

By convention, we use Roman numerals for all of them except factors II/IIa, which are usually designated by the names prothrombin/thrombin, and by factors I/Ia, called fibrinogen/fibrin. Each of the other factors has a regular name too but they are rarely used (you can look these names up if you would like to know).

Cofactors

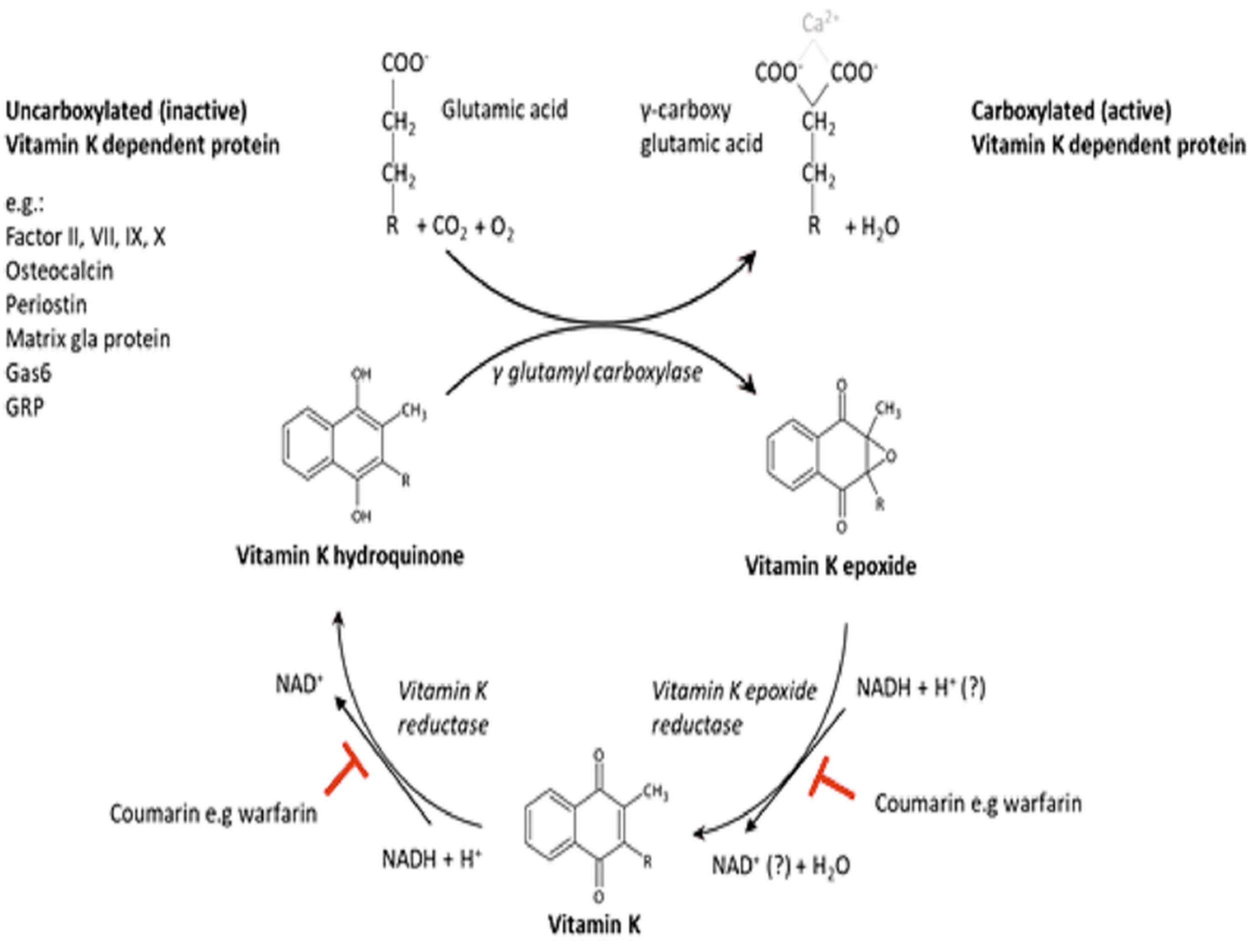
The protein cofactors, whose job it is to augment the coagulation cascade, include tissue factor, Va, and VIIIa.

Besides these cofactors, **ionized calcium** (Ca^{2+}) is also critical for coagulation, as it's needed for the enzymatic conversion of factors II, IX, and X (it is also bound by factor VII but its role there seems less clear).

Clinical Correlation

Severe hypocalcemia has been associated with increased bleeding in trauma patients.

Vitamin K is involved in a different way. It is required for the synthesis of active factors II, VII, IX, and X in the liver, as well as the regulatory proteins C and S. These factors (and some other proteins) contain dicarboxylglutamate which is formed post-translationally by a vitamin K dependent reaction. Regenerating the reduced form of vitamin K happens through a couple of additional reactions, listed as vitamin K epoxide reductase and vitamin K reductase in the figure below. The traditional view is that it is the first of these which can be inhibited by warfarin. However, recent data1 indicates that firstly, the two reactions are performed by the same enzyme (VKORC1) and secondly, there is an uncoupling of the two reactions such that the total inhibition by warfarin is stronger than the inhibition of each part. A different (and equally probable?) interpretation of the data is that warfarin inhibits both reactions to different degrees.



Schematic representation of the Vitamin K cycle. Image credit: Alonso, N., Meinitzer, A., Fritz-Petrin, E., Enko, D., & Herrmann, M. (2022). Role of vitamin K in bone and muscle metabolism. Calcified Tissue International, 112, 1-19. https://doi.org/10.1007/s00223-022-00955-3. CC BY 4.0.

Let us now return to the structure of the other product of the above reaction, the gamma-carboxy glutamic acid. This is necessary for activation of enzymatic activity by Ca^{2+} mentioned above, because the dicarboxylic group directly interacts with calcium.

And for completeness, the above-mentioned proteins (1972CS - and others outside of the clotting cascade) have a specific glutamate that is converted to this dicarboxylic acid to gain that calcium sensitivity.

Clinical Correlation

The anticoagulant drug warfarin interferes with the enzymatic activation of Vitamin K and therefore prevents post-translational modification of these coagulation factors, reducing the activity of the coagulation cascade. This is clinically important. On the flip side, if a patient has taken an overdose of warfarin, giving high dose of vitamin K is able to alleviate at least part of the problem.

Knowledge Check Question

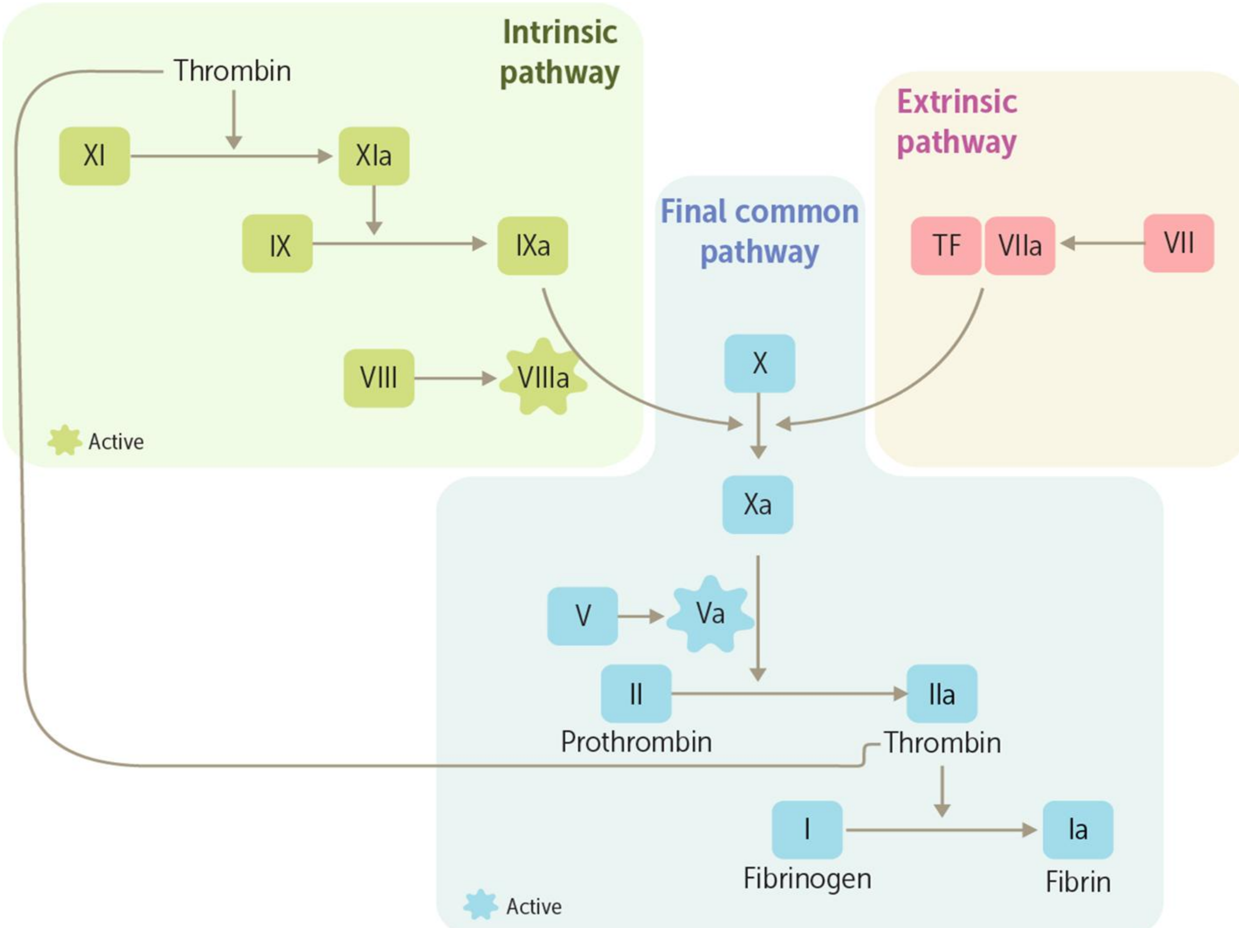
Question: Why is vitamin K important in the coagulation cascade?

Answer: Vitamin K is important in the coagulation cascade because post-translational modification forms gamma-carboxy glutamic acid in factors II, VII, IX, and X (and C and S) in a reaction which requires the reduced form of vitamin K. Mnemonic is 1972CS.

[1] Warfarin alters vitamin K metabolism: a surprising mechanism of VKORC1 uncoupling necessitates an additional reductase - PMC (nih.gov).

What Are the Pathways of the Clotting Cascade?

OK, now we have the cast of characters. Let's see how they interact to produce blood clotting. Looking at the overall structure of the coagulation cascade, notice that it is divided into different pathways—intrinsic, extrinsic, and final common pathways—that are arranged in a kind of Y-like shape joining at factor X.



Coagulation Cascade. Image credit: ScholarRx, used with permission.

Remember, the point of the cascade is to convert fibrinogen to fibrin, which is required for platelet plug formation and clotting. The extrinsic and intrinsic pathways merge at the conversion of factor X to Xa to form the final common pathway, which produces the fibrin.

How are the extrinsic and intrinsic pathways different?

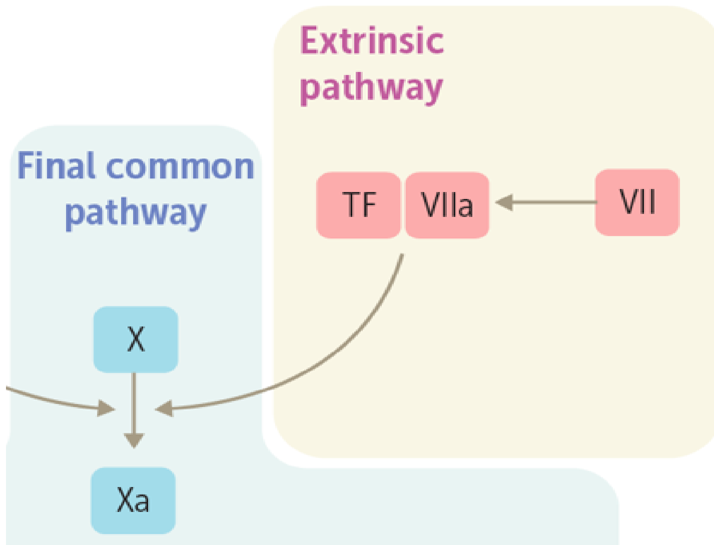
- The extrinsic pathway (extrinsic means "outside") is initiated by injury to blood vessels (extrinsic injury). It kicks things off.
- The intrinsic pathway, in contrast, is initiated in vivo by an intrinsic cause: thrombin (factor IIa) is "intrinsically" produced within the final common pathway. Thrombin then acts to amplify clot formation by initiating the intrinsic pathway.

Let's look at each of these pathways separately, and then we'll see how they fit together.

Extrinsic Pathway

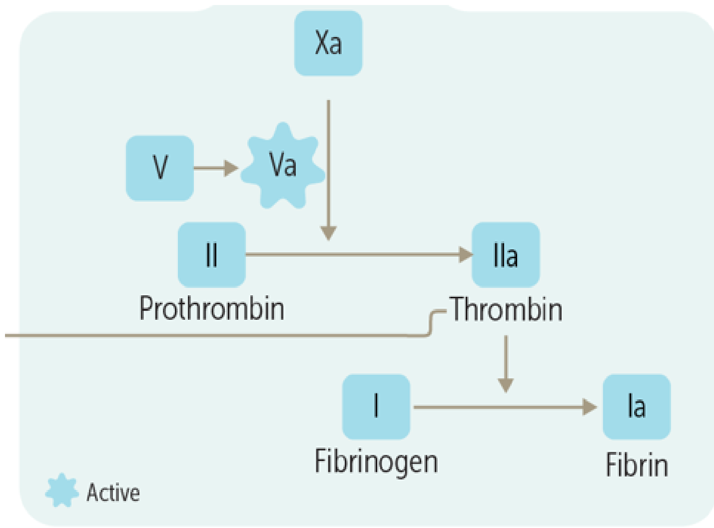
When a vessel is injured, the first thing that happens is activation of the extrinsic pathway, and the purpose of this is to activate factor X to Xa which is first factor in the common pathway. Activation of this pathway happens when the membrane protein **tissue factor** (TF) is exposed to the bloodstream. TF is not exposed to blood under normal resting conditions; otherwise, we'd be constantly coagulating. TF is found in two different places:

- On cells that are normally not in contact with blood (like smooth muscle cells in the subendothelium and fibroblasts surrounding blood vessels). Vascular injury exposes this TF to the blood and initiates the coagulation cascade.
- On endothelial cells and on circulating monocytes when inflammation is present.



Extrinsic Pathway. Image credit: ScholarRx, modified and used with permission.

Once exposed to the blood, TF first binds to factor VII, and this TF/VII complex is activated to TF/VIIa by other serine proteases in the cascade and by neighboring TF/VIIa complexes. The TF/VIIa complex then activates factor X to Xa to kick off the final common pathway, which activates thrombin and makes fibrin, as shown in the below figure, and which will be discussed in a section later in this page.



The final common pathway starting from Xa. Notice that thrombin has roles outside of what is shown here. Image credit: ScholarRx, modified and used with permission.

It turns out that the extrinsic pathway alone is not enough to make the fibrin needed to form a good clot. One reason for this is that as soon as we make a little bit of Xa using the extrinsic pathway, the molecule tissue factor pathway inhibitor (TFPI) shuts down the entire extrinsic pathway. The mechanism for this inhibition is that TFPI binds to Xa, and this complex binds to TFVIIa, meaning that the inhibition acts both on Xa and hinders production of more Xa by TFVIIa. This is one reason that we also need the accompanying intrinsic pathway.

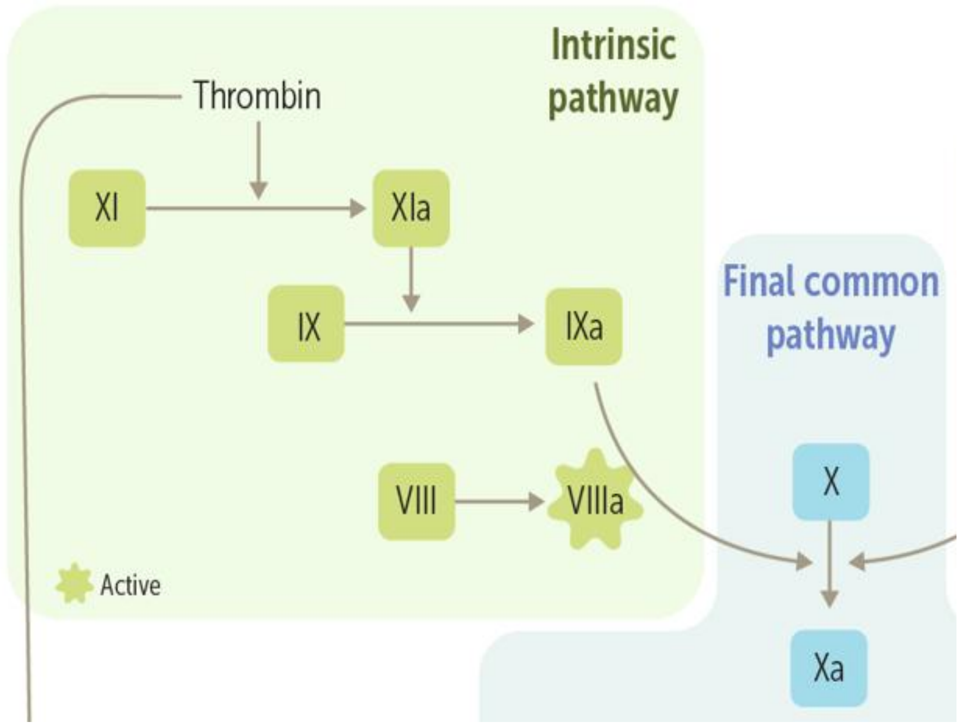
Knowledge Check Question

Question: What is the source of tissue factor (TF)?

Answer: TF is in the subendothelium of the blood vessel and on inflamed monocytes and endothelial cells. When exposed to blood it initiates the extrinsic clotting pathway.

Intrinsic Pathway

As said, **thrombin** generated in the final common pathway is the main activator of the intrinsic pathway in vivo. In the intrinsic pathway, thrombin converts factor XI to XIa, and the cascade rapidly generates factors IXa and VIIIa, which then converts X to Xa and we again activate the final common pathway, as we did in the extrinsic pathway.



Intrinsic pathway to final common pathway. Image credit: ScholarRx, modified and used with permission

A couple of items in the preceding paragraph do need explanation. Firstly, TFPI is found in circulation in a very small concentration (2.5 nM) so it is likely that the intrinsic pathway simply produces more Xa than what can be inhibited by TFPI. Secondly, thrombin has a second role in the intrinsic pathway, it also cleaves factor VIII into VIIIa while this protein is still bound to von Willebrand factor, and once cleaved, VIIIa dissociates from there and binds to IXa in a calcium requiring complex.

A final note is that the function of the intrinsic pathway is measured by the activated partial thromboplastin time (aPTT – reference range 25-40 s) while the extrinsic pathway is measured by prothrombin time (PT – reference range 11-15 s). Bleeding time is not commonly used anymore but it measures the function of the platelets, not the function of the coagulation cascade.

Knowledge Check Question

Question: What is the source of the thrombin which initiates the intrinsic pathway?

Answer: Thrombin is produced in the final common pathway by the complex of Xa and Va acting on prothrombin.

Clinical Correlation - Hemophilia

The two primary types of hemophilia are A and B. Hemophilia A is a result of mutations in the factor VIII gene, whereas hemophilia B is a result of mutations in the factor IX gene. These mutations lead to decreased activity of their respective clotting factors, which then leads to an inability to activate factor X and a consequent defect in fibrin formation. Because of the inability to form fibrin, the clot does not stabilize with the fibrin "glue," and excessive bleeding occurs, especially after trauma.

Most hemophilia cases are inherited, but 20-30% of hemophilia A cases arise from spontaneous mutations in the factor VIII gene.

Hemophilia C is autosomal recessive and caused by deficiency of factor XI, leading to deficient production of factor IXa.

Epidemiology
Both types of hemophilia are rare, with only a combined frequency of about 1 in 5,000 live male births. Of the two types, hemophilia A is about four times more common.

How Do Patients with Hemophilia Present?
The clinical features of hemophilia include easy bruising (ecchymoses), as well as uncontrolled hemorrhage after trauma or surgery.

Parts of the body that are subject to repeated mechanical stresses, such as the joints, often experience "spontaneous" hemorrhages (hemarthrosis) that can ultimately result in joint deformities.

The most commonly affected joints are the knees, ankles, and elbows. Due to recurrent joint bleeding, the surrounding muscles, nerves, and underlying bones all suffer structural damage that leads to chronic pain and impaired mobility. The progressive worsening of the symptoms may eventually lead to joint replacement or in less developed countries to amputation.

Bleeding can also occur from the genitourinary tract, leading to persistent hematuria, as well as the gastrointestinal tract, leading to blood in the stool.
Severity of hemophilia is variable and proportional to the remaining level of factor present. Symptoms are usually detectable if levels (concentration or activity) are less than 40% of normal, and severest symptoms in patients with less than 1% of normal.

Knee deformity. Image credit: Dr. James Heilman, via ScholarRx, used with permission.

Knowledge Check Question

Question: Defects in which coagulation factors cause hemophilia A and B?

Answer: Hemophilia A results from a defect in factor VIII and hemophilia B from a defect in factor IX. (an even rarer hemophilia C is autosomal recessive and caused by deficient factor XI).

You'll notice that some factors such as high-molecular-weight kininogen, prekallikrein, and factor XII are not shown in the Coagulation cascade image, above, as part of the intrinsic pathway. That's because although technically these factors are able to activate the intrinsic pathway, they don't play much of a role in the body. Factor XII, in particular, is only important in forming clots in test tubes. People with deficiencies of these factors (sometimes called contact factors) don't have bleeding disorders but their clotting tests in the test tube may be abnormal!

Knowledge Check Question

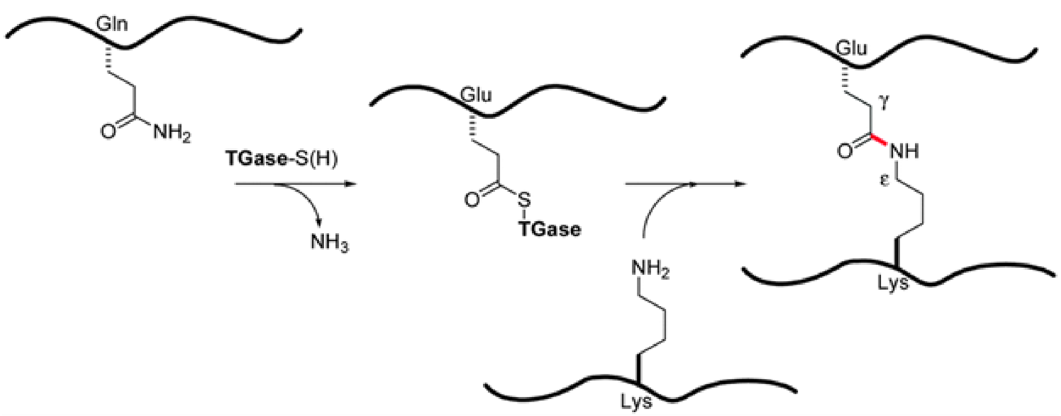
Question: What coagulation factor imbalances might cause clinically apparent bleeding disorders?

Answer: Deficiencies in coagulation factors VIII or IX might cause bleeding disorders, whereas contact factor imbalances (eg, factor XII) do not.

Final Common Pathway

This is the last leg of the cascade and can be activated by either the intrinsic or extrinsic pathway by the conversion of X to Xa. This pathway consists of factors X, V, II (thrombin), and I (fibrinogen). Just like with activation of factor VIII, conversion of V to Va also requires activated thrombin. Factor X, once converted to Xa can complex with Va (again in a calcium containing complex) and this complex converts prothrombin (factor II) to thrombin (IIa), and finally, IIa converts fibrinogen to fibrin. The thrombin circles back to activate the intrinsic pathway and the cofactors as described above. Fibrin then glues together the clot—job done!

There is one last factor (XIII) that you will sometimes hear about. It is not technically part of the cascade, but instead is produced by platelets and acts to stabilize the forming blood clot. This is another factor activated by thrombin; once activated, it forms covalent crosslinks between fibrins in the form of amide bonds between glutamine and lysine.



Transglutaminase (TGase)-mediated protein cross-linking. Image credit: Oteng-Pabi, S. K., Pardin, C., Stoica, M., & Keillor, J. W. (2014). Site-specific protein labelling and immobilization mediated by microbial transglutaminase. Chemical Communications, 50(50), 6604–6606. https://doi.org/10.1039/c4cc00994k. CC BY 3.0.

Knowledge Check Question

Question: At what point in the coagulation cascade do both pathways converge?

Answer: Both pathways converge in the final common pathway, at the activation of factor X in the coagulation cascade.

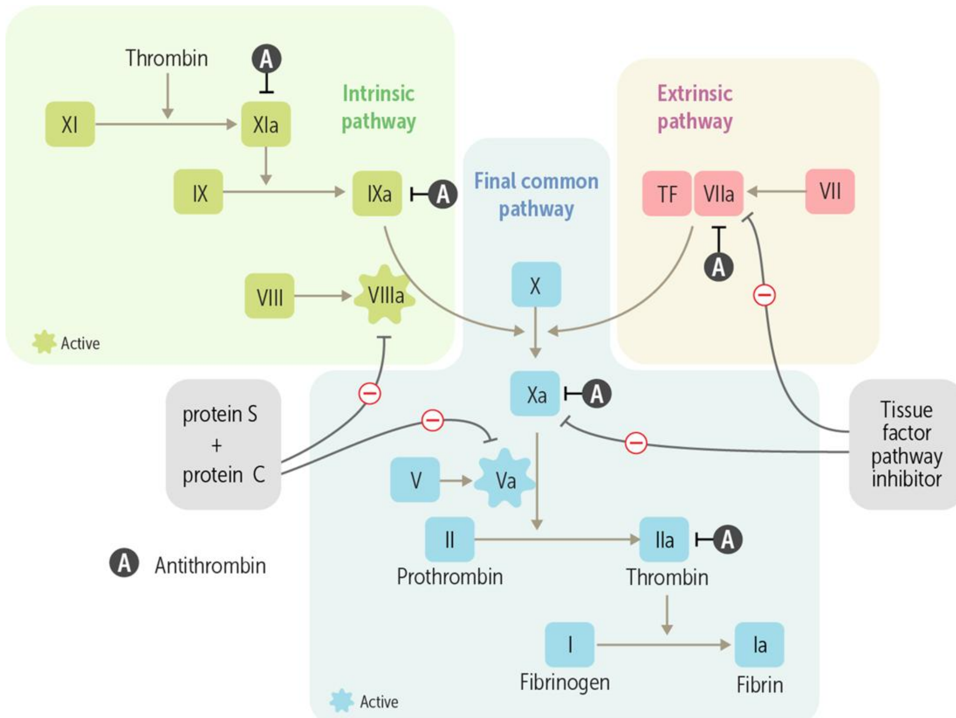
The clotting factors of the coagulation pathway are summarized the table below, along with their cofactors.

Clotting Factors and Cofactors of the Coagulation Pathway

Factor (Selected Common Names)	Description	Cofactors
Factor I (Fibrinogen)	Soluble plasma protein, activated to form fibrin in final common pathway	
Factor II (Prothrombin)	Activation to thrombin required for formation of fibrin (Ia) in final common pathway	Vitamin K, Ca^{++}
Factor III (Tissue Factor)	Released to start extrinsic pathway	
Factor V	Activation required for formation of thrombin (IIa) in final common pathway	
Factor VII	Required for activation of X to Xa in extrinsic pathway	Vitamin K, Ca^{++} , TF
Factor VIII	Required for activation of X to Xa activator in intrinsic pathway	
Factor IX	Required for activation of X to Xa activator in intrinsic pathway	Vitamin K, Ca^{++} , VIIIa
Factor X	Activation required for formation of thrombin (IIa) in final common pathway	Vitamin K, Ca^{++} , Va
Factor XI	Required for the activation of factor IX in intrinsic pathway	
Factor XII	Activation initiates intrinsic pathway in vitro, but minimal action in vivo	

How Does the Body Stop the Coagulation Cascade?

There are several **circulating inhibitors** of the coagulation cascade that the body uses for just this reason, including antithrombin, protein C, protein S, and tissue factor pathway inhibitor. Note that "A" in the figure designates the effect of antithrombin, which has multiple sites of inhibition.



Circulating inhibitors of the coagulation cascade. Image credit: ScholarRx, used with permission.

These inhibitors are clinically important both because genetic deficiencies of them can lead to hypercoagulability (frequent clotting) and because they interact with common anticoagulant drugs like heparin.

Antithrombin

Antithrombin is made in the liver and inhibits many of the clotting factors. It has its strongest effects on **IIa (thrombin) and factor Xa**, although it also inhibits VIIa, IXa, and Xia. Antithrombin forms irreversible bonds with activated coagulation factors, which stops the action of the coagulation factors and clears them from the circulation. You may also hear the term "antithrombin III": this is actually the active form of antithrombin. The literature is moving toward using "antithrombin" alone—but you can use the terms interchangeably.

Clinical Correlation

Antithrombin is activated by the anticoagulant heparin, used in many patients with pathologic blood clotting.

Protein C and Protein S

Protein C is a serine protease while protein S is a non-enzyme cofactor; both are made in the liver. Like other serine proteases in the coagulation cascade, C is made as an inactive zymogen that needs to be cleaved to become active; this cleavage is usually done by thrombin which has been bound and complexed by thrombomodulin, a cell surface molecule that changes the substrate specificity of thrombin to allow recognition of C. Once activated, C complexes with S and they act together to inactivate two important cofactors: VIIIa and Va by cleavage. Factor VIIIa is part of the intrinsic arm of the cascade, and factor Va is part of the final common pathway (see **Circulating inhibitors of the coagulation cascade image above**).

Both proteins C and S contain gamma-carboxy glutamate, just like factors II, VII, IX and X; it is therefore expected that binding of Ca^{2+} is an important part of their function.

Clinical Correlation

Protein C or S genetic deficiencies can lead to abnormal clotting, including deep venous thrombosis and pulmonary embolism.

Knowledge Check Question

Question: How do proteins C and S affect the coagulation cascade?

Answer: Proteins C and S act on both the intrinsic pathway (through factor VIIIa inactivation) and the final common pathway (through factor Va inactivation), thus exerting control over clot formation.

Tissue Factor Pathway Inhibitor

Tissue factor pathway inhibitor (TFPI) is a circulating protease inhibitor that inhibits factor X activation. It does this by both directly inhibiting factor Xa and by binding to and inactivating the tissue factor-factor VIIa complex (see **Circulating inhibitors of the coagulation cascade image above**). The bottom line is that it blocks the initiation of the extrinsic pathway.

Unlike the other inhibitors mentioned above, TFPI is made mostly by vascular endothelial cells, and mostly stays on the cell surface. The molecule is released into the circulation after administration of the anticoagulant drug heparin. This release of endothelial TFPI may contribute to the antithrombotic effects of heparin.

What Is Fibrinolysis?

Besides inhibiting the coagulation cascade, the other main way to keep clotting in check is to break down any small clots that may already have formed, despite the inhibition pathway. This process is called fibrinolysis because it involves chopping up into little fragments the fibrin that's holding the clot together. Without fibrin holding the clot together, the clot simply dissolves.

Fibrinolysis is used by the body in a few different settings:

- **New clot formation**: Although it might be counterintuitive, fibrinolysis is an important part of new clot formation, because it's critical to limit the size and scope of a clot. Almost as soon as a clot is formed, fibrinolytic mechanisms kick in and "remodel" the clot to keep it to an appropriate size.
- **Abnormal clots**: Fibrinolysis is also used to break down clots that have formed abnormally (such as deep venous thrombosis, an abnormal clot that has formed in the deep veins of the leg). Fibrinolytic mechanisms are often able to dissolve these abnormal clots, at least in part, so that blood flow can be restored before there is irreversible damage downstream from the clot.

How does fibrinolysis occur? Fibrin is broken down by an enzyme called **plasmin**, which is formed from an inactive precursor molecule called plasminogen. Circulating **tissue plasminogen activator** (tPA) catalyzes the conversion of plasminogen to plasmin.

Clinical Correlation

tPA can be used as a drug. Recombinant tPA is used as a drug in acute myocardial infarction and strokes to dissolve pathologic clots.

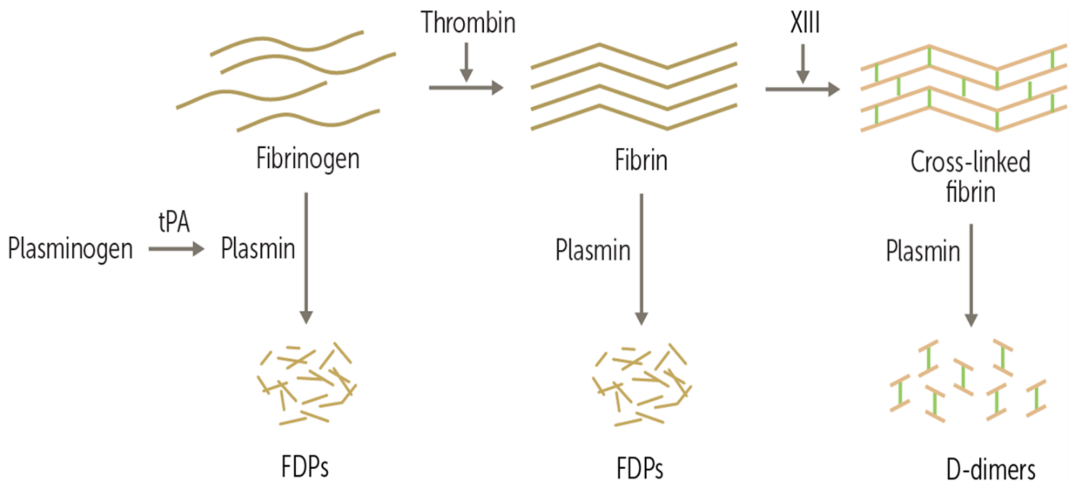
Knowledge Check Question

Question: How does fibrinolysis occur?

Answer: Fibrinolysis occurs as plasminogen is converted to plasmin by tPA. Plasmin then digests fibrin, which breaks down the clots.

Fibrin Degradation Products

Fibrinolysis results in several different fibrin degradation products (FDPs) that can be measured in the blood. This is helpful in diagnosing people who have abnormal clotting (called thrombosis), obstructing veins or arteries.



Fibrinolysis results in several different FDPs. Image credit: ScholarRx, used with permission.

FDPs are just small fragments of fibrin that increase in the blood if we're breaking down a clot. However, measuring FDPs diagnostically as a way of detecting pathologic blood clots is problematic for a couple reasons:

- The test is hyper-sensitive. It will pick up FDPs that are formed by breaking down even tiny, minor, nonpathologic clots.
- FDPs can result from breaking down fibrinogen (see left side of **Fibrinolysis results in several different FDPs, above**), not just fibrin. Therefore, the presence of FDPs doesn't always mean that fibrin as part of a clot is being broken down.

D-Dimers

As an alternative to FDPs, you can also measure a different degradation product called D-dimers (see right side of **Fibrinolysis results in several different FDPs**, above). These are chunks of fibrin that have been crosslinked by factor XIII, so you know they are actually from a real clot (not from fibrinogen, which doesn't get crosslinked by factor XIII). However, there's also a problem with the D-dimer assay. It's the same thing that was wrong with the FDP assay: D-dimers become visible in blood with even minor clotting (like the kind that happens if you sit all day in class and damage tiny capillaries). So, there are a lot of false positives for these tests.

Fortunately, this ultra-sensitivity of both tests can be used to our advantage: if a patient is suspected of having a pathologic thrombus (like a deep venous thrombus or pulmonary embolus) and we do an FDP or D-dimer test and they come back negative, then we can be fairly certain that the patient does not have a pathologic thrombus; if they did, there would be all kinds of D-dimers floating around, and the test would have picked them up. Bottom line: FDP and D-dimer assays are not great for ruling in a clot, but they are great for ruling out a clot. To rule them in, we generally need direct visualization of the clot using ultrasound, CT, or MRI scanning.

Knowledge Check Question

Question: What is the best clinical application of the D-dimer test?

Answer: Because of the high sensitivity of the D-dimer test, the best clinical application is for ruling out pathologic thrombi (if the D-dimer test is normal, you can be very sure that the patient does not have a pathologic thrombus).

Summary

- The coagulation cascade is a series of enzymatic reactions that result in the formation of fibrin, which is important to seal a platelet plug.
- Enzymes (or factors) form the backbone of the coagulation cascade, and they include factors II, VII, IX, X, XI, XII, and XIII.
- There are also cofactors whose job it is to amplify the coagulation cascade; these include ionized calcium, vitamin K, tissue factor, and factors Va and VIIIa.
- The cascade is divided into an extrinsic arm, an intrinsic arm, and a final common pathway.
- When a vessel is damaged, tissue factor is exposed, and the cascade starts along the extrinsic pathway, which activates factor X to start the final common pathway.
- The main components of the extrinsic pathway are tissue factor and factor VII.
- The intrinsic pathway is begun in vivo by thrombin and leads to activation of factor X to start the final common pathway.
- The main components of the intrinsic pathway are factors VIII, IX and XI.
- The final common pathway starts with the formation of factor Xa and culminates by turning fibrinogen (factor I) into fibrin (factor Ia), which then stabilizes the soft platelet plug.